

VAIBHAV GARG

Data Engineer | Data Engineer | AWS Certified (DEA-C01 · AIF-C01) | PL-300 Certified | Python | PySpark | SQL | Snowflake | Power BI | AI/ML

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Post-Study Work Visa (Subclass 485) — 4 years remaining | No sponsorship required | Available Immediately | Open to Sydney · Melbourne · Brisbane · Perth · Adelaide

PROFESSIONAL SUMMARY

Results-driven Software and Data Engineer with 4.5 years of combined professional and applied research experience designing, building, and delivering scalable ETL/ELT pipelines, cloud-native data architectures, and AI-ready data systems using Python, PySpark, SQL, AWS, and Snowflake. Holds three active certifications: AWS Certified Data Engineer – Associate (DEA-C01, 2026), Microsoft Certified Power BI Data Analyst Associate (PL-300, 2026), and AWS Certified AI Practitioner (AIF-C01, 2026). Master of Artificial Intelligence and Machine Learning, University of Adelaide, Australia (GPA 6.6/7.0, 2025). Delivered consistent, measurable outcomes: 30% reduction in pipeline execution time, 5+ TB of operational data processed daily with zero SLA breaches, 95%+ elimination of production data errors, and zero-incident cloud migration. Equally strong in software engineering fundamentals — RESTful APIs, modular code, automated testing, CI/CD — and in applied AI/ML across NLP, computer vision, and predictive modelling. Fully work-authorised in Australia on a Post-Study Work Visa (Subclass 485) with 4 years remaining — no sponsorship required. Available immediately.

CORE SKILLS

Languages & Query: Python (advanced), SQL (advanced), PySpark, Java (basic), Shell Scripting

Data Engineering: ETL/ELT Pipeline Design, Apache Spark, PySpark, Hadoop, Hive, Data Ingestion, Transformation, Data Quality, Data Lineage, Medallion Architecture (Bronze/Silver/Gold), Airflow, dbt

AWS (DEA-C01 Certified): S3, Glue, Lambda, EC2, Redshift, Kinesis, Step Functions, IAM, CloudWatch, EMR, Athena, DMS, API Gateway, SNS, SQS

AI & Cloud (AIF-C01 Certified): AWS AI/ML services, Generative AI foundations, AI-ready data infrastructure, intelligent automation, LLMs, AI Practitioner certified

Cloud & Warehousing: Snowflake, GCP, Azure (basics), Data Lakes, Dimensional Modelling, Multi-cloud Architecture, Apache Kafka

BI & Reporting (PL-300 Certified): Power BI (Microsoft Certified — PL-300), DAX, data modelling, dashboard design, KPI reporting, operational analytics

Python Ecosystem: Pandas, NumPy, Scikit-learn, SQLAlchemy, FastAPI, Flask, pytest, Matplotlib, Seaborn

ML & AI: Scikit-learn, Regression, Classification, Clustering (K-Means), Time-Series, NLP, Computer Vision (YOLOv8, OpenCV), ML Model Deployment, feature engineering

Software Engineering: RESTful API Design, Microservices, OOP, Secure Coding, Code Review, Performance Optimisation, Spring Boot

DevOps & CI/CD: Git, GitHub Actions, Docker, AWS CodePipeline, pytest — unit & integration testing, Secrets Management

Databases: Snowflake, Redshift, MySQL, SQLite, DynamoDB — schema design, migrations, query optimisation

PROFESSIONAL EXPERIENCE

Software Engineer — Data & Analytics | Impetus Technologies, India

Aug 2021 – Dec 2023

Python · PySpark · SQL · AWS (S3, Glue, Lambda, Redshift, Kinesis, Step Functions, CloudWatch, IAM) · Snowflake · GCP · Airflow · dbt · Power BI · REST APIs · Docker · pytest · Git · Agile

- **Achieved 30% faster ETL pipeline execution processing 5+ TB daily** — re-architected distributed PySpark pipelines on AWS Glue using Medallion architecture; redesigned partition strategy, eliminated shuffle-heavy joins, parallelised Step Functions-orchestrated job chains with zero SLA breaches post-deployment.
- **Reduced production data errors by 95%+** — built Python data-quality validation frameworks enforcing schema integrity, null/duplicate detection, and referential consistency at every AWS Glue ingestion stage; zero downstream rework requests after implementation.
- **Delivered zero-incident multi-cloud migration** — migrated legacy on-premise databases to AWS (S3, Glue, Lambda) and Snowflake; designed dimensional schemas, built Python reconciliation scripts, and onboarded BI teams onto Power BI automated dashboards within 8 weeks.
- **Processed thousands of daily API requests with zero security incidents** — developed RESTful APIs (Spring Boot) with role-based authentication, IAM policies, input validation, and structured error handling powering an ML loan eligibility scoring engine.
- **Cut pipeline failure detection from hours to under 15 minutes** — configured CloudWatch custom metrics, alarms, and log groups; maintained runbooks and SOPs in Git for incident response and knowledge transfer.
- **Delivered ML and AI-driven data solutions across Retail, Finance, and Operations** — built classification, regression, time-series, and clustering models using Python and Scikit-learn; designed AI-ready data pipelines and feature engineering layers for cross-functional client teams.
- **Maintained delivery quality and engineering standards across Agile sprints** — conducted code reviews, managed feature branches, automated test pipelines with pytest, facilitated sprint ceremonies, and prepared weekly leadership reports for international stakeholders.

Console Operator (Responsible Person) | OTR Group — Viva Energy, Adelaide

Apr 2024 – Present

Power BI · Operational Analytics · Incident Management · SLA Compliance

- **Applied Power BI in a live production environment** — monitored real-time operational KPIs, identified anomalies, applied structured resolution workflows, and maintained SLA compliance with full documentation alongside postgraduate study.

- **Managed operational incidents under pressure** — prioritised competing issues, applied SOPs, and maintained compliance standards; demonstrates structured problem-solving and accountability in a regulated Australian enterprise environment.

KEY PROJECTS

Pelmorex — Multi-Cloud Data Pipeline Migration | *PySpark · AWS · Snowflake · GCP · Python · Airflow · dbt* | *Proprietary*

- **Delivered analytics data 2x faster, 30% reduction in manual operations** — migrated North American weather data platform from legacy on-premise to multi-cloud AWS + Snowflake + GCP; Medallion architecture, PySpark optimisation, Python orchestration — zero SLA misses.
- **Built cross-environment data quality and reconciliation layer** — Python validation enforcing integrity across AWS, Snowflake, and GCP; automated reconciliation reports with signed-off stakeholder delivery for every pipeline run.

ATM Transaction Analytics Pipeline | *PySpark · Python · SQL · Docker · GitHub Actions · pytest* | *vaibhavgarg003.github.io*

- **12/12 pytest tests passing, full CI/CD pipeline** — production PySpark pipeline processing 10,000 transactions; time-bucket classification, high-value flagging, Parquet partitioned output, Dockerised with GitHub Actions automation.

Football Tactical Formation Detection — AI/ML | *YOLOv8 · DeepSORT · OpenCV · K-Means · Python* | *vaibhavgarg003.github.io*

- **Validated against SoccerNet-v2** — end-to-end AI/ML pipeline; YOLOv8 detection, HSV segmentation, DeepSORT tracking, K-Means formation classification across a full 90-minute match. Demonstrates applied AI/ML data product engineering.

ETL Pipeline — Country Data Engineering | *Python · Pandas · SQL · SQLAlchemy · Docker · pytest* | *vaibhavgarg003.github.io*

- **15/15 tests passing, 195 countries, live REST API** — production ETL pipeline; API extraction, Pandas transformation, SQLite load via SQLAlchemy; Dockerised, structured logging, secrets management.

EDUCATION

Master of Artificial Intelligence & Machine Learning — University of Adelaide, Australia*Jan 2024 – Dec 2025*

GPA: 6.6/7.0 | Adelaide Graduate Award (Nov 2025) — leadership, professional excellence, cross-cultural collaboration

- Coursework: Deep learning, NLP, computer vision, reinforcement learning, time-series, ML model deployment, Generative AI, distributed data systems.

Bachelor of Computer Science (B.E./B.Tech) — Rajasthan Technical University 2017 – 2021
80% aggregate

CERTIFICATIONS

AWS Certified Data Engineer – Associate (DEA-C01) — Amazon Web Services, 2026

Microsoft Certified: Power BI Data Analyst Associate (PL-300) — Microsoft, 2026

AWS Certified AI Practitioner (AIF-C01) — Amazon Web Services, 2026

PROFESSIONAL REFEREES

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